

SmartCube ESS Series

RTHY(80~125)KWP3-RTL(197~233)kwh

Versatile Application

- Ideal for factories, hotels, and farms with DC-coupled solar-storage
- On/off-grid switching < 20 ms with genset support

Safe & Reliable

- CATL cells ensure safety and long life
- Fire safety: alarms, valves, smoke / CO sensors, UL9540A certified (Cell, Module)

Smart Energy

- Supports self-consumption, peak shifting, time-of-use, and battery backup
- Integrated BMS / EMS for flexibility

High Performance

- Smart air-cooling for efficiency
- Long-lasting battery cycle life

Easy Operation

- Compact design saves space
- Cloud control with 24 / 7 monitoring



Outdoor Battery Cabinet Parameters

Technical Parameters	RTL197kwh		RTL215kwh		RTL233kwh	
Battery Type	LFP					
Battery Module Capacity	17.92 kWh					
Number of Modules	11		12		13	
Total Battery Capacity	197.12 kWh		215 kWh		233 kWh	
Nominal Voltage	704 V		768 V		832 V	
Operating Voltage Range	627 ~ 792 V		682 ~ 864 V		741 ~ 936 V	
Charge / Discharge Rate	0.5 C					
DoD	90%					
General Parameters						
Dimensions (W x H x D)	1300 x 2380 x 1442 mm					
Weight	< 2.5 T					
Installation Site	Outdoor					
IP Protection	IP54					
Anti Corrosion Level	C4					
Operation Humidity	5% ~ 95% (No Condensing)					
Operation Temperature	-30°C ~ +50°C					
Max. Operation Altitude	≤ 3000 m					
Communication Port	Ethernet; CAN					
Communication Protocol	CAN; TCP					
Cooling Method	Air Conditioner					
Certificates	UN38.3; MSDS; IEC 62040; IEC 62477; IEC 62619; IEC 63056; IEC 61000-6-2/4					

Hybrid Inverter Parameters

Product Specifications	RTHY80KWP3		RTHY100KWP3	RTHY125KWP3
PV Side				
MPPT Voltage Range	250 ~ 950 V (Max. 1000 V)			
MPPT Rated Voltage	720 V			
MPPT Rated Voltage (Full Load)	450 ~ 800 V	550 ~ 800 V	700 ~ 800 V	
Max. PV Power	160 kWp	200 kWp	250 kWp	
Number of MPPT / Strings per MPPT	8 / 2			
Max. Current per MPPT	45 A			
Battery Side				
Battery Voltage Range	250 ~ 900 V			
Battery Rated Voltage Range	250 ~ 800 V			
Max. DC Current	160 A (80 A x 2)			
Max. DC Power	88 kW	110 kW	125 kW	
Number of DC Inputs	2			
AC Side (On Grid)				
Nominal AC Output Power	80 kW	100 kW	125 kW	
Max. AC Output Power	88 kVA	110 kVA	125 kVA	
Rated AC Current	116 A	144 A	181 A	
Max. AC Current	240 A	250 A	250 A	
AC Rated Voltage / Voltage Range	230 / 400 Vac; 220 / 380 Vac; 3L+PE+N; -15% ~ +10%			
Nominal Grid Frequency / Frequency Range	50 Hz / 60 Hz (±5 Hz)			
THDi	< 3% (100% Load)			
Adjustable PF Range	-0.8 (Lagging) ~ 0.8 (Leading)			
Backup Output				
Nominal AC Voltage	230 / 400 V; 220 / 380 Vac; 3L+PE+N			
THDv	< 3% (Rated Power)			
Nominal Grid Frequency / Frequency Range	50 Hz / 60 Hz			
Nominal AC Output Power	80 kW	100 kW	125 kW	
Max. AC Output Power (Single-phase)	50 kW			
Max. Output Current	220 A			
Genset Input				
Max. Input Current	220 A			
Efficiency				
Max. Efficiency	98%			
Protection				
Reverse DC Connection Protection	Yes			
Anti-Islanding Protection	Yes			
Over-Temperature Protection	Yes			
Grid Monitoring / Earthing Fault Detection	Yes			
Insulation Monitoring	Yes			
DC / AC Surge Protection	DC Type II; AC Type II			
AFCI	Optional			
General Parameters				
Dimensions (W x H x D)	635 x 1100 x 330 mm			
Weight	> 95 kg			
Topology	Transformerless			
IP Protection	IP65			
Operation Temperature Range	-25 ~ 60°C (> 45°C Derating)			
Operation Humidity Range	0 ~ 100% (No Condensing)			
Cooling Method	Intelligent Air Cooling			
Max. Operation Altitude	4000 m (> 3000 m Derating)			
Communication Port	RS-485 / CAN			